

SIXTH SEMESTER (CUCBCSS-UG) DEGREE EXAMINATION, MARCH 2022**Mathematics****MAT 6B 12—NUMBER THEORY AND LINEAR ALGEBRA****(2014 to 2018 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 30****Maximum : 30 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 30.
2. The candidate should check that the question paper supplied to him/her contains all the 30 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 6B 12—NUMBER THEORY AND LINEAR ALGEBRA

(Multiple Choice Questions for SDE Candidates)

1. A subset W of a vector space V is a subspace if :
 - (A) The sum of elements of W belongs to W .
 - (B) The sum and scalar multiples of elements of W belongs to W .
 - (C) The scalar multiples of elements of W belongs to W .
 - (D) The sum of elements of V belongs to W .
2. The intersection of any set of subspaces of a vector space V is :
 - (A) Not a subspace of V .
 - (B) A subspace of V .
 - (C) Need not be a subspace of V .
 - (D) A proper subspace of V .
3. The union of any two subspaces of a vector space V is :
 - (A) A subspace of V .
 - (B) Not a subspace of V .
 - (C) Need not be a subspace of V .
 - (D) A proper subspace of V .
4. If a vector space V is of dimension n then every subset containing more than n elements is :
 - (A) Linearly independent.
 - (B) Linearly dependent.
 - (C) Neither linearly independent nor dependent.
 - (D) A basis.
5. If W is a subspace of a finite dimensional vector space V then :
 - (A) $\dim V = \dim W$.
 - (B) $\dim V \leq \dim W$.
 - (C) $\dim V \geq \dim W$.
 - (D) $\dim V \neq \dim W$.
6. The subset $\{(1, 1), (1, -1)\}$ of real vector space $\mathbb{R}^2$ is :
 - (A) Linearly independent only.
 - (B) Linearly dependent.
 - (C) Neither linearly independent nor dependent.
 - (D) A basis.

7. A subset of a linearly independent set is :

- (A) Linearly independent.
- (B) Linearly dependent.
- (C) Neither linearly independent nor dependent.
- (D) Need not be linearly independent.

8. If the map $f: V \rightarrow W$ is linear. Then the kernel or null space of f is :

- (A) $f^{-1}(W)$.
- (B) $f^{-1}(\{0_W\})$.
- (C) $f^{-1}(\{0_V\})$.
- (D) $f^{-1}(W)$.

9. Suppose a and b are integers with $a \neq 0$, then $a \mid b$ if :

- (A) $a = bc$, c is some integer.
- (B) $b = ac$, c is some integer.
- (C) $c = ab$, c is some integer.
- (D) $ab = 1$.

10. If a and b are two non-zero integers and $\gcd(a, b) = d$, then d is the greatest positive integer with :

- (A) $d \mid a$ and $d \mid b$.
- (B) $d \mid a$ only.
- (C) $d \mid b$ only.
- (D) $a \mid d$ and $b \mid d$.

11. $\gcd(-5, 5) = \underline{\hspace{2cm}}$.

- (A) 3.
- (B) 1.
- (C) 5.
- (D) -5.

12. $\gcd(-8, 36) = \underline{\hspace{2cm}}$.

- (A) 36.
- (B) -8.
- (C) 8.
- (D) 4.

13. Let a and b be integers, not both zero. Then a and b are relatively prime if and only if there exists integers x and y such that :

- (A) $1 = ax + by$.
- (B) $2 = ax + by$.
- (C) $ab = ax + by$.
- (D) $a - b = ax + by$.

14. If a and b are non-zero integers with $a \mid b$, then $\gcd(a, b) = \underline{\hspace{2cm}}$.

- (A) $|a|$.
- (B) b .
- (C) ab .
- (D) a .

15. Two integers a and b , not both of which are zero, are said to be relatively prime if :
- (A) $\gcd(a, b) = a$. (B) $a \mid b$.
 (C) $\gcd(a, b) = 1$. (D) $b \mid a$.
16. If a, b, c are any integers with $a \mid bc$ and $\gcd(a, b) = 1$, then :
- (A) $b \mid a$. (B) $a \mid c$.
 (C) $c \mid a$. (D) $c \mid b$.
17. If a is an odd integer then $\gcd(3a, 3a + 2) = \text{-----}$.
- (A) 3. (B) 5.
 (C) 1. (D) 2.
18. Every positive integer $n > 1$ can be expressed as a product of :
- (A) Composite numbers. (B) Prime numbers.
 (C) Even numbers. (D) Odd numbers.
19. If p is a prime and $p \mid ab$, then :
- (A) $p \mid a$ only. (B) $p \mid b$ only.
 (C) $p \mid a$ or $p \mid b$. (D) $p \mid a$ and $p \mid b$.
20. The *Sieve of Eratosthenes* is used for finding :
- (A) All primes below a given integer.
 (B) All even numbers below a given integer.
 (C) All odd numbers below a given integer.
 (D) All composite numbers below a given integer.
21. The International Standard Book Number (ISBN) consists of nine digits $a_1 a_2, \dots, a_9$ followed by a tenth check digit a_{10} , which satisfies :
- (A) $a_{10} \equiv \sum_{k=1}^9 k a_k \pmod{9}$. (B) $a_{10} \equiv \sum_{k=1}^9 k a_k \pmod{11}$.
 (C) $a_{10} \equiv \sum_{k=1}^9 k a_k \pmod{10}$. (D) $a_{10} \equiv \sum_{k=1}^9 k a_k \pmod{7}$.
22. Any *palindrome* with even number of digits is divisible by :
- (A) 5. (B) 2.
 (C) 11. (D) 12.

23. The number of primes of the form $n^2 - 2$ is :

- (A) Finite. (B) Infinite.
(C) 1729. (D) 1.

24. If p and q are distinct prime with $a^p \equiv a \pmod{q}$ and $a^q \equiv a \pmod{p}$, then :

- (A) $a^{p+q} \equiv a \pmod{pq}$. (B) $a^{pq} \equiv a \pmod{pq}$.
(C) $a^{p-q} \equiv a \pmod{pq}$. (D) $a^{p/q} \equiv a \pmod{pq}$.

25. A composite number n for which $a^n \equiv a \pmod{n}$ is called :

- (A) A pseudoprime. (B) A prime.
(C) A pseudoprime to the base a . (D) An absolute pseudoprime.

26. If p is a prime, then $(p-1)! + 1$ is :

- (A) a multiple of p . (B) a multiple of $p-1$.
(C) a multiple of $p+1$. (D) a multiple of p^2 .

27. If n and r are positive integers with $1 \leq r < n$, then the binomial coefficient $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ is :

- (A) An integer. (B) A prime.
(C) Irrational. (D) An even integer.

28. For $n > 2$, $\phi(n)$ is :

- (A) An odd integer. (B) An even integer.
(C) Irrational. (D) Prime.

29. $\phi(360) = \underline{\hspace{2cm}}$.

- (A) 96. (B) 98.
(C) 86. (D) 90.

30. For any integer a , $a^{37} \equiv \underline{\hspace{2cm}} \pmod{1729}$.

- (A) a . (B) $2a$.
(C) a^2 . (D) $a-1$.

SIXTH SEMESTER (CUCBCSS-UG) DEGREE EXAMINATION, MARCH 2022

Mathematics

MAT 6B 12—NUMBER THEORY AND LINEAR ALGEBRA

(2014 to 2018 Admissions)

Time : Three Hours

Maximum : 120 Marks

Section A*Answer all questions.**Each question carries 1 mark.*

1. State general version of the Division Algorithm.
2. Find lcm (306, 657).
3. State the Fundamental Theorem of Arithmetic.
4. Explain the method of Sieve of Eratosthenes.
5. Define Pseudoprime. Give an example of a Pseudoprime number.
6. Find $\sigma(12)$.
7. Define Euler's Phi Function.
8. Prove that the set $X = \{(x, 0) : x \in \mathbb{R}\}$ is a subspace of the vector space $\mathbb{R}^2$.
9. Define basis of a vector space.
10. Check whether the map $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $f(x, y, z) = (x + y, z, 0)$ is linear. Justify your claim.
11. Define the rank and nullity of the linear map.
12. For $(x, y) \in \mathbb{R}^2$, let $S = \{(x, y)\}$. Find $\langle S \rangle$.

(12 × 1 = 12 marks)

Section B*Answer any ten questions.**Each question carries 4 marks.*

13. Show that square of any odd integer is of the form $8k + 1$.
14. If $a \mid c$ and $b \mid c$, with $\gcd(a, b) = 1$, prove that $ab \mid c$.
15. Use the Euclidean Algorithm to obtain integers x and y satisfying $\gcd(24, 138) = 24x + 138y$.
16. Determine all solutions in the positive integers of the Diophantine equation $18x + 5y = 48$.

Turn over

17. Prove that the number $\sqrt{2}$ is irrational.
18. For arbitrary integers a and b , prove that $a \equiv b \pmod{n}$ if and only if a and b leave the same non-negative remainder when divided by n .
19. Use the binary exponential algorithm to compute $5^{110} \pmod{131}$.
20. Solve the linear congruence $18x \equiv 1 \pmod{42}$.
21. If p and q are distinct primes with $a^p \equiv a \pmod{q}$ and $a^q \equiv a \pmod{p}$, then prove that $a^{pq} \equiv a \pmod{pq}$.
22. Prove that intersection of two subspaces of vector space V over a field is again a subspace.
23. Let $A = \text{Span}\{(1, 2, 0, 1), (-1, 1, 1, 1)\}$ and $B = \text{Span}\{(0, 0, 1, 1), (2, 2, 2, 2)\}$ be two subspaces of $\mathbb{R}^4$. Determine $A \cap B$ and compute its dimension.
24. If V is a vector space over the set $\mathbb{C}$ of complex numbers of dimension n , prove that V can be regarded as a vector space over $\mathbb{R}$ of dimension $2n$.
25. Let $f: V \rightarrow W$ be linear. Prove that if Y is subspace of W then $f^{-1}(Y)$ is a subspace of V .
26. Let V be a vector space of dimension $n \geq 1$ over a field F . Prove that V is isomorphic to the vector space F^n .

(10 × 4 = 40 marks)

Section C

*Answer any six questions.
Each question carries 7 marks.*

27. Given integers a and b , not both of which are zero, prove that there exists integers x and y such that $\gcd(a, b) = ax + by$.
28. Prove that the linear Diophantine equation $ax + by = c$ has an integer solution if and only if $d \mid c$ where $d = \gcd(a, b)$.
29. Prove that there are infinite number of primes.
30. Using Chinese Remainder Theorem, solve the system of congruences $x \equiv 5 \pmod{11}, x \equiv 14 \pmod{29}, x \equiv 15 \pmod{31}$.
31. If p is a prime, then prove that $(p-1)! \equiv -1 \pmod{p}$.
32. Prove that a non-empty subset S of a vector space V is a basis of V if and only if every element of V can be expressed in a unique way as a linear combination of elements of S .
33. Let $f, g, h: \mathbb{R} \rightarrow \mathbb{R}$ be the mappings given by $f(x) = \cos^2 x, g(x) = \sin^2 x, h(x) = \cos 2x$. Consider the subspace of $\text{Diff}(\mathbb{R}, \mathbb{R})$ given by $W = \text{Span}\{f, g, h\}$. Find a basis for W .